1 I

1.1 (1)

Using the Heisenberg equation of motion, we have:

$$\frac{d\sigma^2}{dt} = \frac{1}{i\hbar}[\sigma^2, H],$$

where

$$\sigma^2 = \sigma_x^2 + \sigma_y^2 + \sigma_z^2.$$

According to the commutation relation between the Pauli matrices:

$$[\sigma_i, \sigma_j] = 2i \sum_k \epsilon_{ijk}\sigma_k,$$

we obtain:

$$\begin{aligned}
[\sigma^2, \sigma_j] &= \sum_i [\sigma_i^2, \sigma_j] \\
&= \sum_i (\sigma_i[\sigma_i, \sigma_j] + [\sigma_i, \sigma_j]\sigma_i) \\
&= 2i \sum_{i,k} \epsilon_{ijk}(\sigma_i\sigma_k + \sigma_k\sigma_i) \\
&= 0.
\end{aligned}$$

Here, we have used the fact that different Pauli matrices anticommute.

Therefore:

$$\begin{aligned}
[\sigma^2, H] &= -\hbar\omega[\sigma^2, n_x\sigma_x + n_z\sigma_z] \\
&= 0.
\end{aligned}$$

Thus:

$$\boxed{\frac{d\sigma^2}{dt} = 0},$$

which tells us that σ^2 is conserved.

1.2 (2)

In the basis of $|\uparrow\rangle$ and $|\downarrow\rangle$, the Hamiltonian can be expressed as:

$$H = -\hbar\omega \begin{pmatrix} n_z & n_x \\ n_x & -n_z \end{pmatrix}.$$

The eigenvalues can be obtained by solving the characteristic equation:

$$\begin{vmatrix} -\hbar\omega n_z - \lambda & -\hbar\omega n_x \\ -\hbar\omega n_x & \hbar\omega n_z - \lambda \end{vmatrix} = 0.$$

Expanding the determinant, we obtain:

$$\lambda^2 - \hbar^2\omega^2(n_x^2 + n_z^2) = 0.$$

Since $n_x^2 + n_z^2 = 1$, the two eigenvalues are:

$$\boxed{\lambda = \pm\hbar\omega}.$$

1.3 (3)

For the ground state with eigenvalue $-\hbar\omega$, we can obtain:

$$-\hbar\omega \begin{pmatrix} n_z & n_x \\ n_x & -n_z \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -\hbar\omega \begin{pmatrix} a \\ b \end{pmatrix}.$$

From the first row, we have:

$$(n_z - 1)a + n_x b = 0.$$

Comparing with the form provided in the problem, we obtain:

$$\boxed{\tan\theta = \frac{b}{a} = \frac{1 - n_z}{n_x}}.$$

2 II

2.1 (4)

Substituting

$$|\psi(t)\rangle = c_1(t)|1\rangle + c_2(t)|2\rangle$$

into the time-dependent Schrödinger equation (TDSE):

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle,$$

and noticing that

$$H|1\rangle = -\hbar\omega|1\rangle, \qquad H|2\rangle = \hbar\omega|2\rangle,$$

we obtain:

$$i\hbar\left(\dot{c}_1|1\rangle + \dot{c}_2|2\rangle\right) = -\hbar\omega c_1|1\rangle + \hbar\omega c_2|2\rangle.$$

Comparing the coefficients of the two eigenstates, we have:

$$\boxed{\begin{aligned} &\frac{d}{dt}c_1(t) = i\omega c_1(t), \\ &\frac{d}{dt}c_2(t) = -i\omega c_2(t). \end{aligned}}$$

2.2 (5)

The initial state is:

$$|\psi(0)\rangle = \cos\theta|1\rangle - \sin\theta|2\rangle.$$

Therefore:

$$c_1(0) = \cos\theta, \qquad c_2(0) = -\sin\theta.$$

Solving the differential equations obtained in (4), we obtain:

$$\begin{aligned} c_1(t) &= \cos\theta\, e^{i\omega t}, \\ c_2(t) &= -\sin\theta\, e^{-i\omega t}. \end{aligned}$$

Thus, the state at time t is:

$$\boxed{|\psi(t)\rangle = \cos\theta\, e^{i\omega t}|1\rangle - \sin\theta\, e^{-i\omega t}|2\rangle}.$$

2.3 (6)

Substituting the expressions of $|1\rangle$ and $|2\rangle$ into the result of (5), we obtain:

$$\begin{aligned}|\psi(t)\rangle = &\cos\theta e^{i\omega t}(\cos\theta|\uparrow\rangle + \sin\theta|\downarrow\rangle)\\ &-\sin\theta e^{-i\omega t}(-\sin\theta|\uparrow\rangle + \cos\theta|\downarrow\rangle).\end{aligned}$$

Rearranging the coefficients and using Euler's formula, we have:

$$\boxed{\begin{aligned}|\psi(t)\rangle = &(\cos\omega t + i\cos 2\theta\sin\omega t)|\uparrow\rangle\\ &+(i\sin 2\theta\sin\omega t)|\downarrow\rangle.\end{aligned}}$$

Using this state, the expectation value of σ_z is:

$$\begin{aligned}\langle\sigma_z\rangle &= |\cos\omega t + i\cos 2\theta\sin\omega t|^2\\ &\quad -|i\sin 2\theta\sin\omega t|^2\\ &= \cos^2\omega t + \cos^2 2\theta\sin^2\omega t\\ &\quad -\sin^2 2\theta\sin^2\omega t\\ &= 1 - 2\sin^2 2\theta\sin^2\omega t.\end{aligned}$$

Therefore:

$$\boxed{\langle\sigma_z\rangle = 1 - 2\sin^2 2\theta\sin^2\omega t}.$$

3 III

3.1 (7)

According to the definitions of the instantaneous energy eigenstates:

$$\begin{aligned}|1(t)\rangle &= \cos\theta(t)|\uparrow\rangle + \sin\theta(t)|\downarrow\rangle,\\ |2(t)\rangle &= -\sin\theta(t)|\uparrow\rangle + \cos\theta(t)|\downarrow\rangle,\end{aligned}$$

where $\theta(t) = \Omega t/2$, we obtain:

$$\begin{aligned}\frac{d}{dt}|1(t)\rangle &= \frac{\Omega}{2}(-\sin\theta(t)|\uparrow\rangle + \cos\theta(t)|\downarrow\rangle)\\ &= \frac{\Omega}{2}|2(t)\rangle,\end{aligned}$$

and similarly:

$$\begin{aligned}\frac{d}{dt}|2(t)\rangle &= -\frac{\Omega}{2}(\cos\theta(t)|\uparrow\rangle + \sin\theta(t)|\downarrow\rangle)\\ &= -\frac{\Omega}{2}|1(t)\rangle.\end{aligned}$$

Thus:

$$\boxed{\begin{aligned}\frac{d}{dt}|1(t)\rangle &= \frac{\Omega}{2}|2(t)\rangle,\\ \frac{d}{dt}|2(t)\rangle &= -\frac{\Omega}{2}|1(t)\rangle.\end{aligned}}$$

3.2 (8)

Substituting

$$|\psi(t)\rangle = a_1(t)|1(t)\rangle + a_2(t)|2(t)\rangle$$

into the TDSE, we obtain:

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = i\hbar\Big[\dot{a}_1|1(t)\rangle + a_1\frac{d}{dt}|1(t)\rangle + \dot{a}_2|2(t)\rangle + a_2\frac{d}{dt}|2(t)\rangle\Big].$$

Using the results of (7) and the instantaneous eigenvalues:

$$H(t)|1(t)\rangle = -\hbar\omega|1(t)\rangle, \qquad H(t)|2(t)\rangle = \hbar\omega|2(t)\rangle,$$

we obtain:

$$i\hbar\left(\dot{a}_1 - \frac{\Omega}{2}a_2\right) = -\hbar\omega a_1,$$
$$i\hbar\left(\dot{a}_2 + \frac{\Omega}{2}a_1\right) = \hbar\omega a_2.$$

Therefore:

$$\boxed{\begin{aligned}\dot{a}_1 - \frac{\Omega}{2}a_2 &= i\omega a_1,\\ \dot{a}_2 + \frac{\Omega}{2}a_1 &= -i\omega a_2.\end{aligned}}$$

3.3 (9)

At $t = 0$, we have:

$$|1(0)\rangle = |\uparrow\rangle, \qquad |2(0)\rangle = |\downarrow\rangle.$$

Since the initial state is $|\psi(0)\rangle = |\uparrow\rangle$, the initial conditions are:

$$a_1(0) = 1, \qquad a_2(0) = 0.$$

When $\Omega/\omega \to 0$, which means $\Omega \ll \omega$, the coupling terms proportional to Ω are negligible compared with the energy splitting. Thus:

$$\dot{a}_1 \approx i\omega a_1,$$
$$\dot{a}_2 \approx -i\omega a_2.$$

With the initial conditions, we obtain:

$$a_1(t) \approx e^{i\omega t}, \qquad a_2(t) \approx 0.$$

Therefore, the system approximately remains in the instantaneous ground state throughout the rotation. At the final time $t = \pi/\Omega$, we have:

$$|1(\pi/\Omega)\rangle = |\downarrow\rangle.$$

Thus, the final state is:

$$\boxed{|\psi(\pi/\Omega)\rangle \approx e^{i\omega\pi/\Omega}|\downarrow\rangle}.$$

Ignoring the overall phase, the answer is $\boxed{(え)}$.

On the other hand, when $\omega/\Omega \to 0$, which means $\Omega \gg \omega$, the energy terms become negligible compared with the coupling terms. Equation (8) becomes:

$$\dot{a}_1 \approx \frac{\Omega}{2} a_2,$$
$$\dot{a}_2 \approx -\frac{\Omega}{2} a_1.$$

Solving these equations under the same initial conditions, we obtain:

$$a_1(t) = \cos\frac{\Omega t}{2},$$
$$a_2(t) = -\sin\frac{\Omega t}{2}.$$

At the final time $t = \pi/\Omega$:

$$a_1 = 0, \qquad a_2 = -1.$$

Since

$$|2(\pi/\Omega)\rangle = -|\uparrow\rangle,$$

the final state is:

$$\boxed{|\psi(\pi/\Omega)\rangle = -|2(\pi/\Omega)\rangle = |\uparrow\rangle}.$$

Thus, the answer is $\boxed{(あ)}$.